

1. Introduction

This assignment demonstrates the deployment and management of an application using Kubernetes. It covers key concepts such as Pods, ReplicaSets, Deployments, Services, Rolling Updates, and Health Probes.

2. Application Overview

A frontend application was created with two versions:

- **Version 1:** Displays *"Version 1"*
- **Version 2:** Displays *"Version 2"* with UI changes (background color)

This helps demonstrate Kubernetes rolling updates.

3. Docker Image Creation

Two Docker images were created using Nginx:

- smrjo/k8s-app:v1
- smrjo/k8s-app:v2

Commands Used:

```
docker build -t smrjo/k8s-app:v1 -f docker/Dockerfile.v1 .  
docker build -t smrjo/k8s-app:v2 -f docker/Dockerfile.v2 .
```

```
docker push smrjo/k8s-app:v1  
docker push smrjo/k8s-app:v2
```

4. Kubernetes YAML Files

The following Kubernetes resources were created:

- Pod (pod.yml)
- ReplicaSet (rs.yml)
- Deployment (deployment.yml)
- Service (service.yml)

Each file defines how the application runs inside Kubernetes.

5. Deployment Execution

The application was deployed using:

```
kubectl apply -f assignment/  
kubectl get pods
```

The service was exposed using:

```
minikube service myapp-service
```

6. Rolling Update

The application was updated from Version 1 to Version 2 using a rolling update:

```
kubectl set image deployment/myapp-deployment myapp=smrjo/k8s-app:v2  
kubectl get pods -w
```

During this process:

- Old pods were terminated
 - New pods were created
 - No downtime occurred
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7. Probes Implementation

Three types of probes were configured in the deployment:

- **Liveness Probe:** Ensures container is running properly
- **Readiness Probe:** Ensures pod is ready to serve traffic
- **Startup Probe:** Handles slow startup scenarios

These probes improve reliability and fault detection.

8. Verification Commands

kubectl describe deployment myapp-deployment
kubectl rollout history deployment myapp-deployment

These commands were used to verify deployment status and probe configuration.

